

number of minor ones) fails under the modification of some assumption, or what is called “perturbation,” when you change one parameter, or take a parameter heretofore assumed by the theory to be fixed and stable, and make it random. We call this “randomization” in the jargon. This is called the study of model error and examination of the consequences of such changes (my official academic specialty is now model error or “model risk”). For instance, if a model used for risk assumes that the type of randomness under consideration is from Mediocristan, it will ignore large deviations and encourage the building of a lot of risk that ignores large deviations; accordingly, risk management will be faulty. Hence the metaphor of “sitting on a barrel of dynamite” I used concerning Fannie Mae (now bust).

For another example of egregious model error, take the notion of comparative advantage supposedly discovered by Ricardo and behind the wheels of globalization. The idea is that countries should focus, as a consultant would say, on “what they do best” (more exactly, on where they are missing the smallest number of opportunities); so one country should specialize in wine and the other in clothes, although one of them might be better at both. But do some perturbations and alternative scenarios: consider what would happen to the country specializing in wine if the price of wine fluctuated. Just a simple perturbation around this assumption (say, considering that the price of wine is random, and can experience Extremistan-style variations) makes one reach a conclusion the opposite of Ricardo’s. Mother Nature does not like overspecialization, as it limits evolution and weakens the animals.

This also explains why I found current ideas on globalization (such as those promoted by the journalist Thomas Friedman) one step too naïve, and too dangerous for society—unless one takes into account side effects. Globalization might give the appearance of efficiency, but the operating leverage and the degrees of interaction between parts will cause small cracks in one spot to percolate through the entire system. The result would be like a brain experiencing an epileptic seizure from too many cells firing at the same time. Consider that our brain, a well-functioning complex system, is not “globalized,” or, at least, not naïvely “globalized.”

The same idea applies to debt—it makes you fragile, very fragile under perturbations, particularly when we switch from the assumption of Mediocristan to that of Extremistan. We currently learn in business schools to engage in borrowing (by the same professors who teach the Gaussian bell curve, that Great Intellectual Fraud, among other pseudosciences), against all historical traditions, when all Mediterranean